

Introductory lecture: Artificial Intelligence, Machine Learning, and Deep Learning

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Objectives

- Practical knowledge about Machine Learning (ML) applications
- Working with online ML platforms/frameworks, e.g., Google Colab, Keras, TensorFlow, PyTorch
- Designing and implementing applications by using various ML algorithms, techniques, and methodologies
- Evaluating a machine learning model by means of real-world datasets

Introduction

- Basic concepts in Artificial Intelligence, Machine Learning (ML), and Deep Learning (DL).
- Notable applications of ML/DL in different domains.
- How a machine learning algorithm learns from data (extract patterns).
- Ongoing research issues.

Artificial Intelligence

- The capability of machines to make decisions which could be comprehended as intelligent in humans.
- *"AI refers to systems that display intelligent behaviour by analysing their environment and taking action – with some degree of autonomy – to achieve specific goals."*

Philip Boucher, Artificial intelligence: How does it work, why does it matter, and what can we do about it?

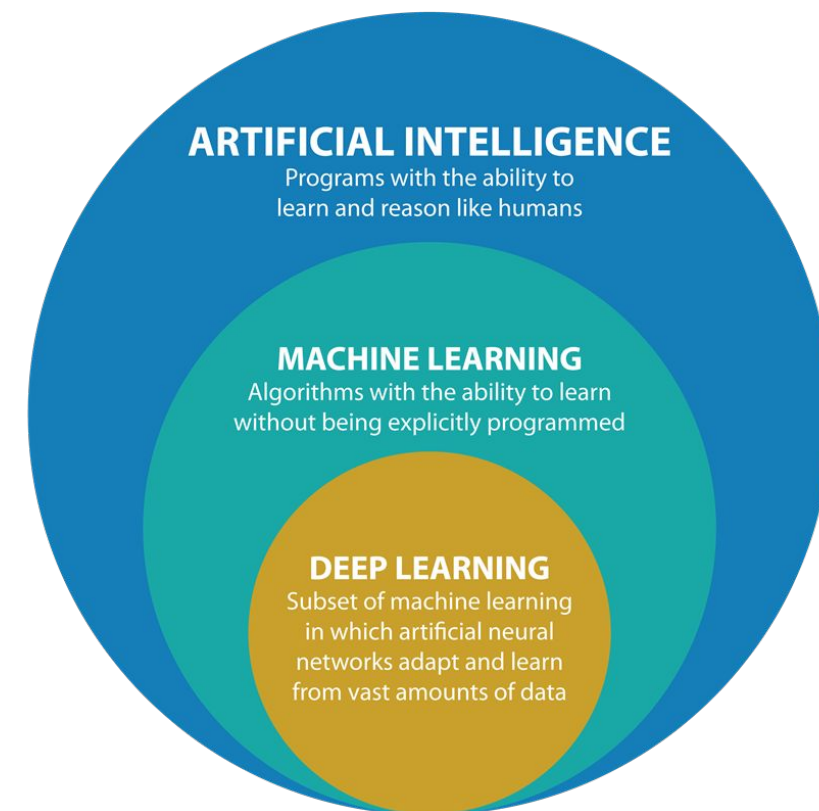


Image source: <https://www.enerbrain.com/de/why-deep-learning-is-important-for-enerbrain/>

Artificial Intelligence: Expert systems

- A set of rules is used to represent knowledge, which can then be executed by computers

```

If the patient has fever
    Prescribe drug X
If the patient is coughing
    Prescribe drug Y
Else
    Send patient home
    
```

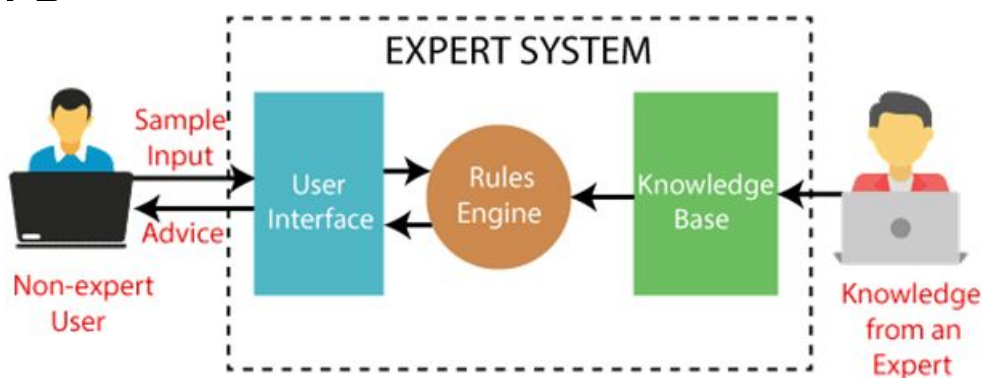


Image source: <https://www.javatpoint.com/expert-systems-in-artificial-intelligence>

- **Mycin**: Finding the bacteria that causes infections.
- **Dendral**: Detect unknown organic molecules using their mass spectra, and knowledge base of chemistry.

Machine Learning

- Algorithms that allow computers to learn from data without being explicitly coded.
- Extract meaningful patterns from data using supervised and unsupervised algorithms.

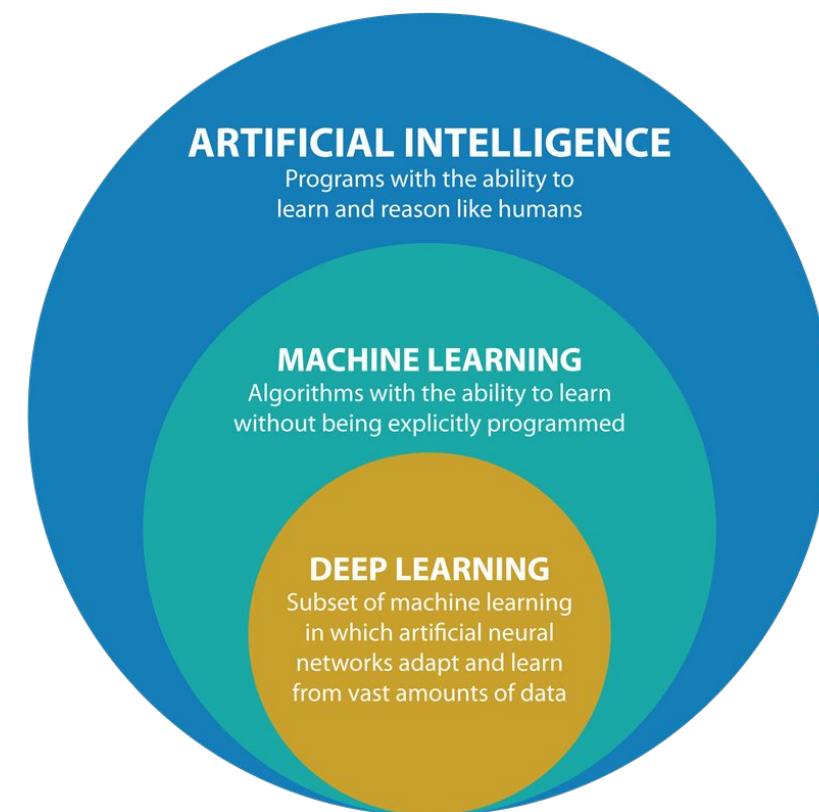


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Machine Learning Algorithms

- Linear regression: Modeling the relationship between a dependent variable and one or more independent variables.
- Logistic Regression: Binary and multi-class classification problems, logistic regression models the probability of a binary outcome.
- Decision Trees: A hierarchical tree-like structure used for both classification and regression tasks, where decisions are made based on feature values.

Machine Learning Algorithms (2)

- Random Forest: Combining multiple decision trees and combines their predictions for improved accuracy and generalization (Ensemble learning).
- Support Vector Machines (SVM): Finding the optimal hyperplane to separate data points into different classes.
- Naive Bayes: A probabilistic algorithm based on Bayes' theorem, often used for classification problems, particularly in text classification.

Machine Learning Algorithms (3)

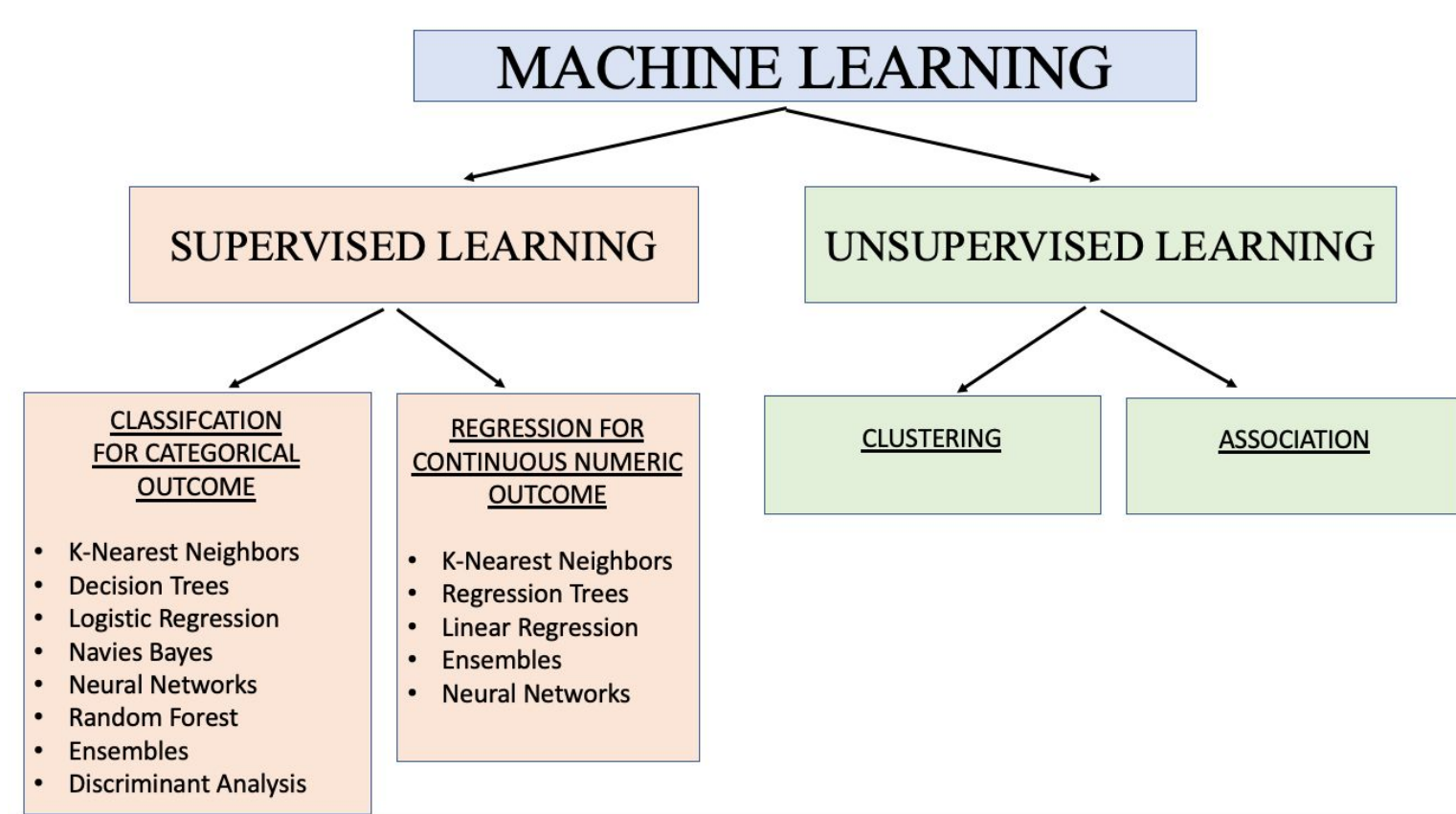


Image source: <https://medium.com/@ooemma83/how-to-identify-supervised-and-unsupervised-machine-learning-models-7707973096f7>

Machine Learning Cheatsheets

- A condensed and handy reference guide, providing key information, tips, and formulas related to machine learning concepts and algorithms.
- Quick references for both beginners and experienced practitioners.
- Helping practitioners quickly access important information during their work or studies.
- They can save time and help reinforce important concepts and techniques in the field of machine learning.

Machine Learning Cheatsheet (2)

Algorithm	Description	Applications	Advantages	Disadvantages
Linear Regression	A simple algorithm that models a linear relationship between inputs and a continuous numerical output variable	1. Stock Price Prediction 2. Predicting housing prices 3. Predicting customer lifetime value	1. Explainable method 2. Interpretable results by its output coefficient 3. Faster to train than other machine learning models	1. Assumes linearity between inputs and output 2. Sensitive to outliers 3. Can underfit with small, high-dimensional data

Image source: <https://www.datacamp.com/cheat-sheet/machine-learning-cheat-sheet>

Deep Learning

- A branch of ML, the use of neural networks with several layers to capture features in data.
- Each layer is used to learn a specific set of features.

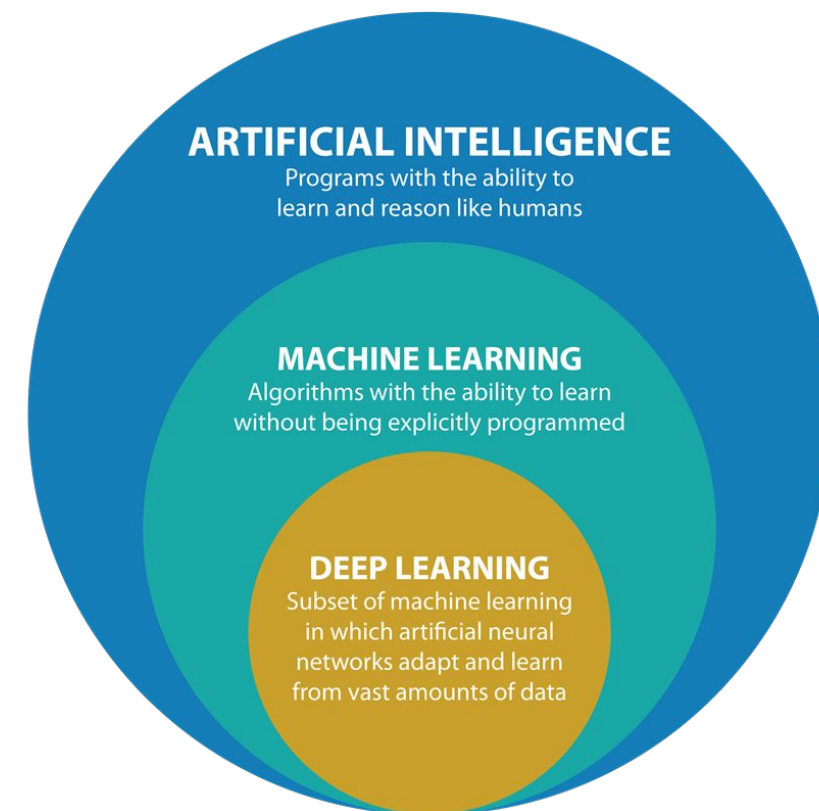


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AI, ML, and DL

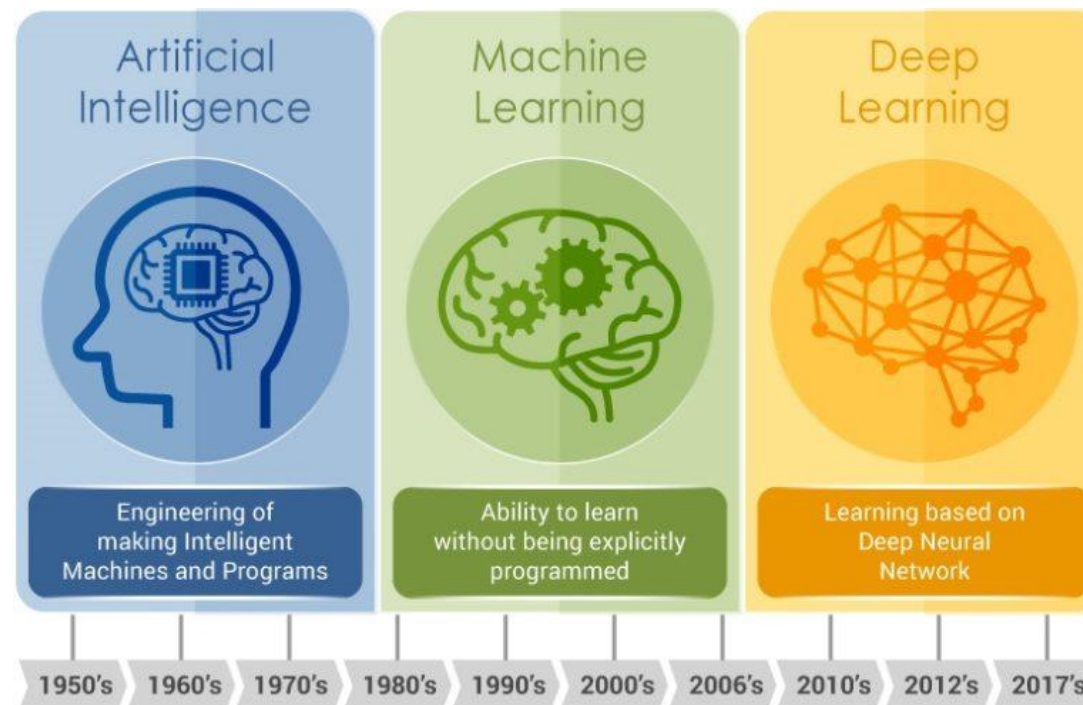


Image source:

<https://www.viatech.com/en/2018/05/history-of-artificial-intelligence/>

- 1950s to 2000s: The development of different ML algorithms.
- 2022: The proliferation of deep learning algorithms being widely used across several domains.

Classification of ML algorithms

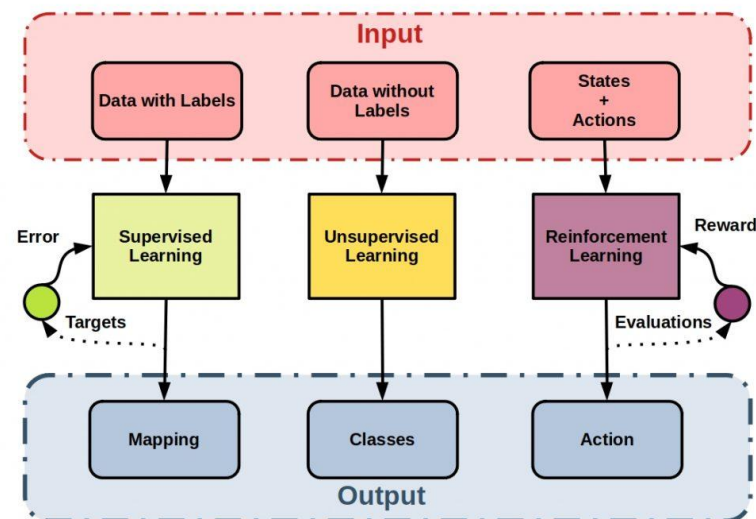


Image source: <https://starship-knowledge.com/supervised-vs-unsupervised-vs-reinforcement>

- Supervised learning: Using labeled data to train models.
- Unsupervised learning: No labeled data is needed.
- Reinforcement learning: Intelligent agents perform actions so as to maximize cumulative reward.

Classification of ML algorithms

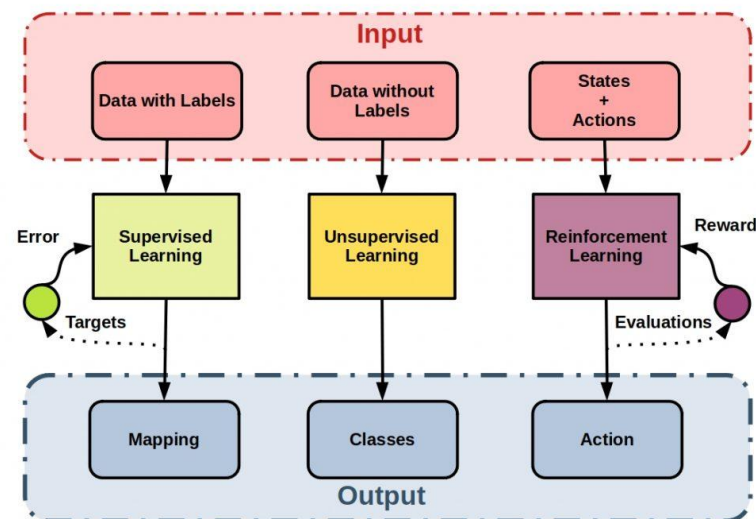


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- **Supervised learning:** Using labeled data to train models.
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Unsupervised learning

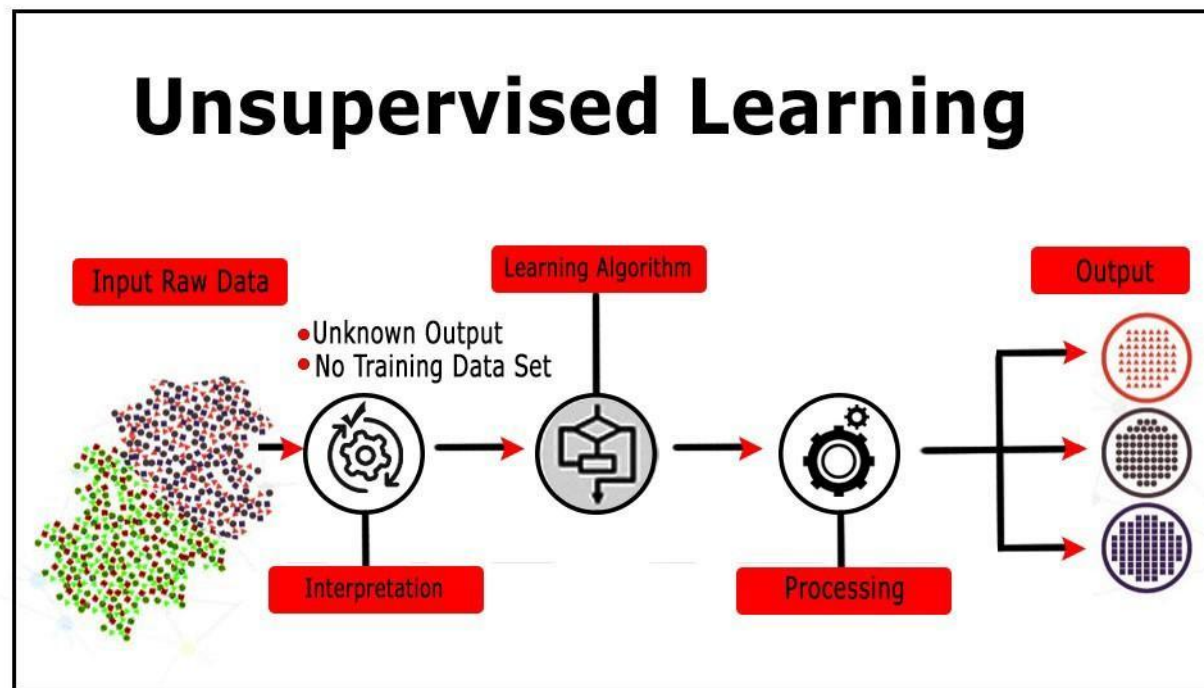


Image source: <https://medium.datadriveninvestor.com/supervised-and-unsupervised-learning-7281050992a0>

- Automatic extraction of patterns from data by analyzing the similarity among instances.
- Application: Clustering.

Unsupervised learning

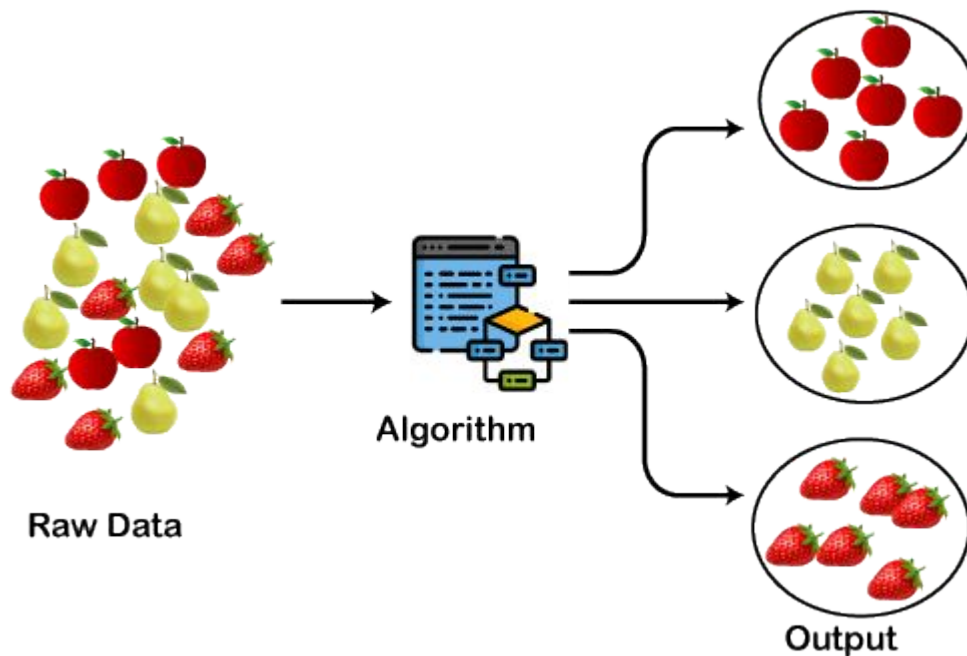
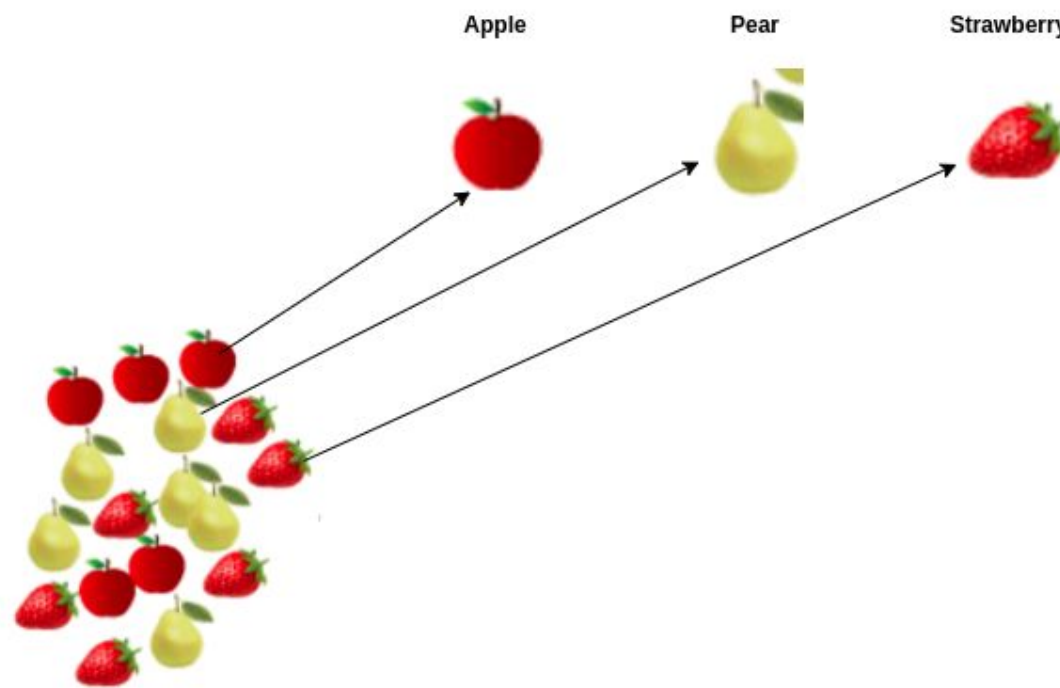


Image source: <https://www.javatpoint.com/clustering-in-machine-learning>

- Clustering: the process of organizing objects into groups of similar objects.
- A proximity (or distance) measure to allow one to assess if elements to be clustered are mutually similar or dissimilar.

Supervised learning



- Classification: We know in advance the categories.
- The model learns from the existing features to assign new object to a specific category.

Introduction

- Supervised learning: One of the fundamental paradigms in machine learning.
- An algorithm learns from labeled training data to make predictions or decisions without being explicitly programmed.
- Supervised learning forms the foundation of many real-world machine learning applications.
- A powerful approach for tasks where you have access to labeled data and want to automate decision-making or make predictions based on that data.

Introduction (2)

- The primary goal of supervised learning is to learn a mapping or relationship between the input data and the output labels.
- The model aims to make accurate predictions or classifications for new, unseen data based on what it has learned from the training dataset.
- Labeled data: A dataset consisting of input-output pairs. Each data point in the dataset includes both the input (features) and the corresponding desired output (label or target).
- Labels are given by humans through manual annotation.

Characteristics

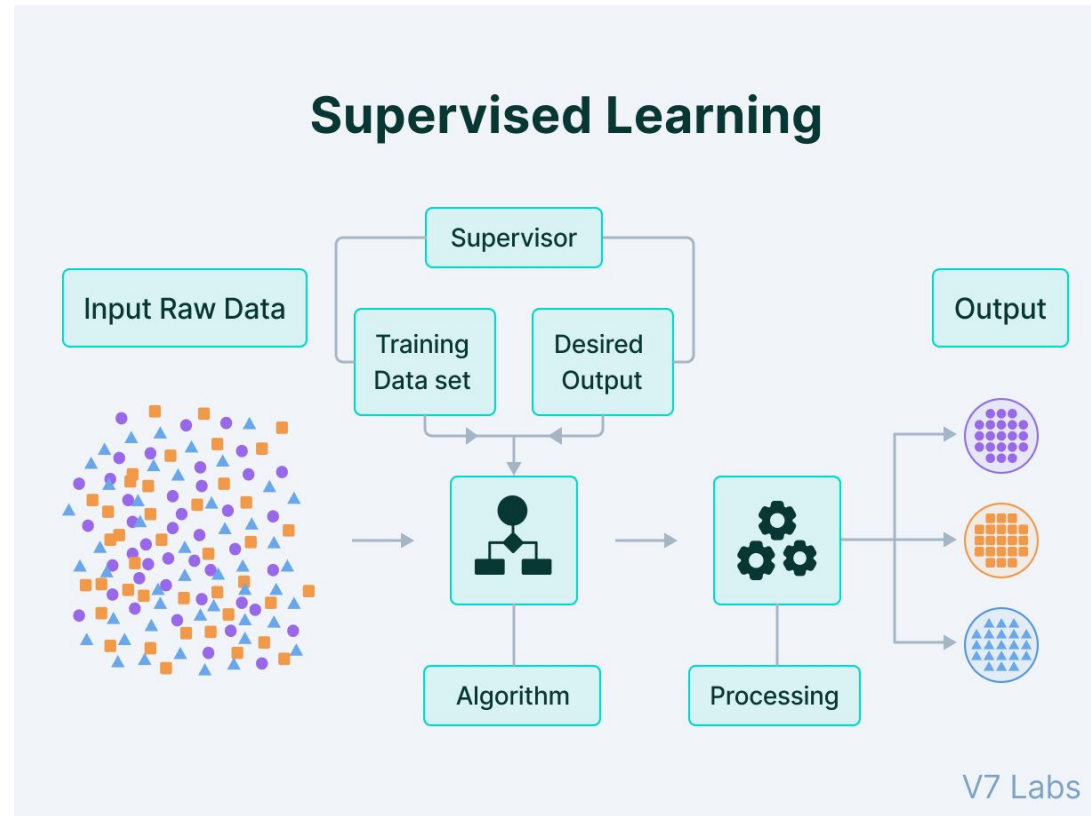


Image source: <https://www.v7labs.com/blog/supervised-vs-unsupervised-learning>

- Trained with labeled data
- Main applications: Classification and regression

Characteristics (2)

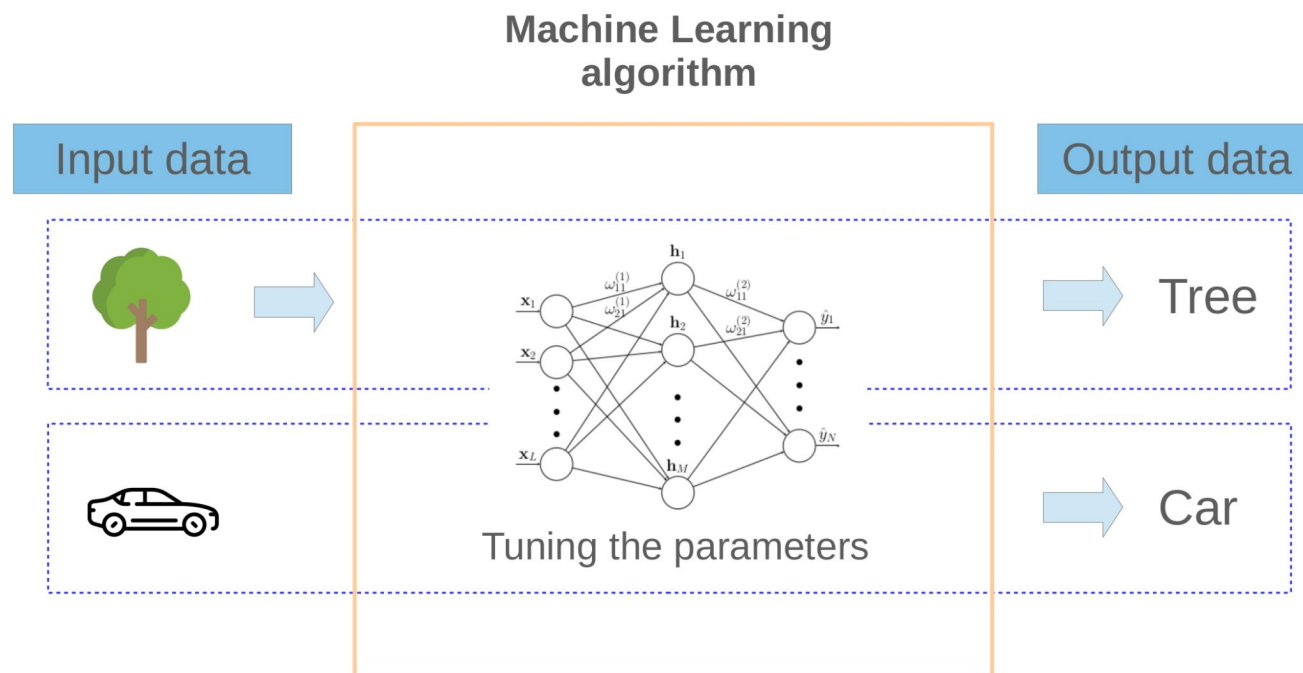
- Training phase: The algorithm uses the labeled training data to adjust its internal parameters and build a model that can make predictions.
- The learning algorithm seeks to minimize the difference between its predictions and the actual labels.
- Testing Phase: After training, the model is tested on a separate dataset (the test set) to evaluate its performance and assess its ability to generalize to new, unseen data.

How to train a Machine Learning classifier

— The learning process

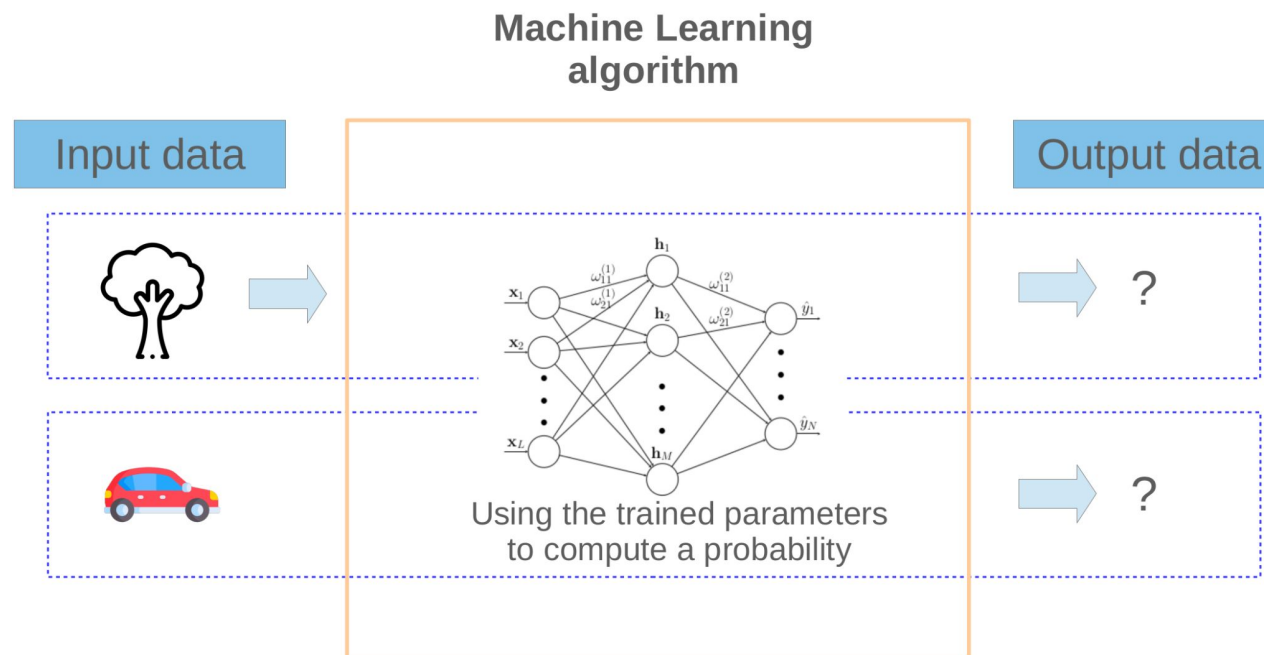
- Learning is actually the process of modifying the weights so that we can produce the desired outputs given the inputs.
- The more (good) input data you feed the network, the better accuracy you will obtain.

Training phase



- Feature: Input data (documents, images).
- Label: Given by humans.

Testing phase



- Feature: Input data (documents, images).
- Label: The algorithm tries to compute the probability, and give a label.

Train a dog

- Commands: "sit down", "shake hand".
- You press him, or grab his hand to instruct the corresponding action.
- Deployment: you speak out the command, and the dog does what he was taught.

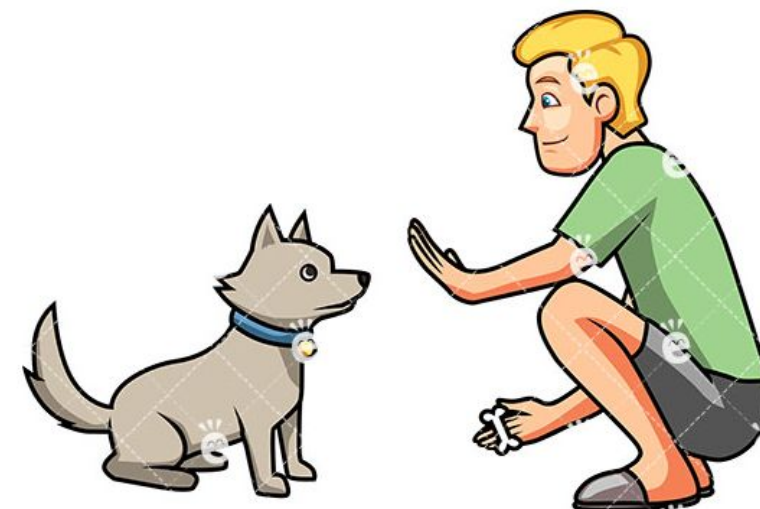
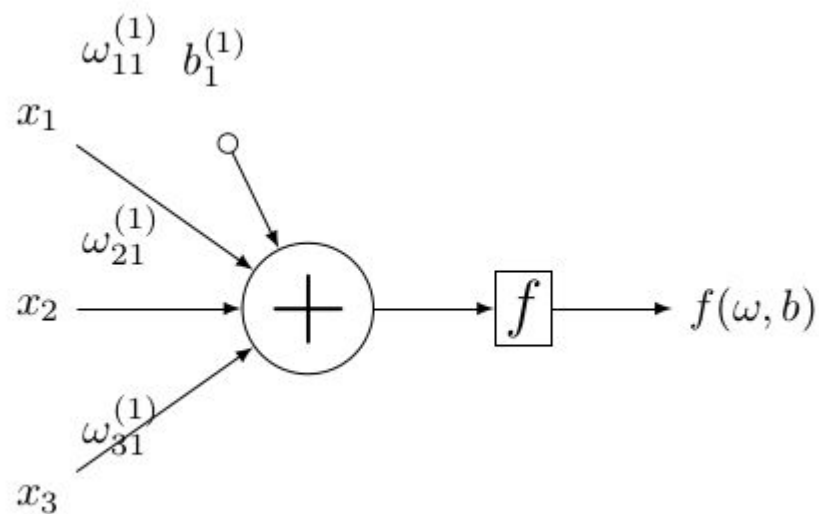


Image source: <https://friendlystock.com/product/man-training-his-dog-to-stay-giving-treat/>

Neural Network: Perceptron

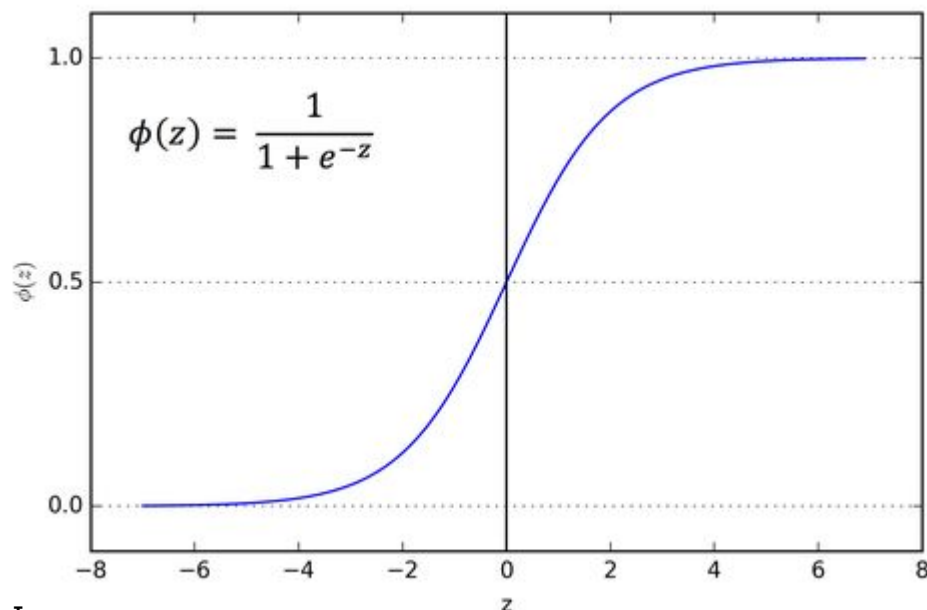


(a) A simple perceptron with 3 inputs

$$output = f\left(\sum_{i=1}^3 \omega_{i1}^{(1)} \cdot x_i + b^{(1)}\right)$$

- A decision making unit, taking into consideration different inputs

Activation: The sigmoid function

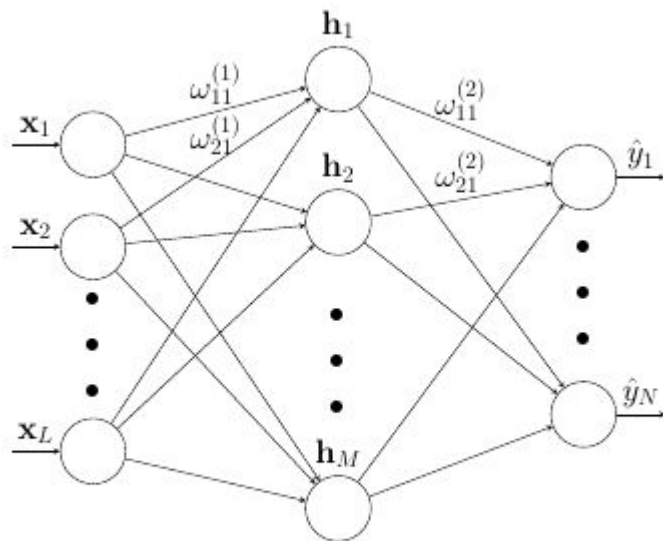


$$y = f(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + \exp(-\sum_j w_j x_j - b)}$$

Image source:
<https://towardsdatascience.com/activation-functions-neural-networks-1cbd9f8d91d6>

- Small changes in the weights (w) and biases (b) will trigger a small change in the output.

Feed-forward Neural Network



(b) A three-layer neural network

$$C(w, b) = \frac{1}{2n} \sum_x \|y(x) - (w.x + b)\|^2$$

$$\Delta y \approx \sum_j \frac{\partial y}{\partial w_j} \Delta w_j + \frac{\partial y}{\partial b} \Delta b$$

- Learning is actually the process of modifying the weights so that we can produce the desired outputs given the inputs.
- The more (good) input data you feed the network, the better accuracy you will obtain.

Learning

- Find a set of **weights and biases** (w, b) that minimize the cost function between the predicted values and the real ones using Stochastic Gradient Descent (SGD)



- Δy is a linear function of Δw and Δb

$$\Delta y \approx \sum_j \frac{\partial y}{\partial w_j} \Delta w_j + \frac{\partial y}{\partial b} \Delta b$$

Minimizing the cost function

- Find a set of weights that minimize the cost function between the predicted value and the real one using Stochastic Gradient Descent (SGD).

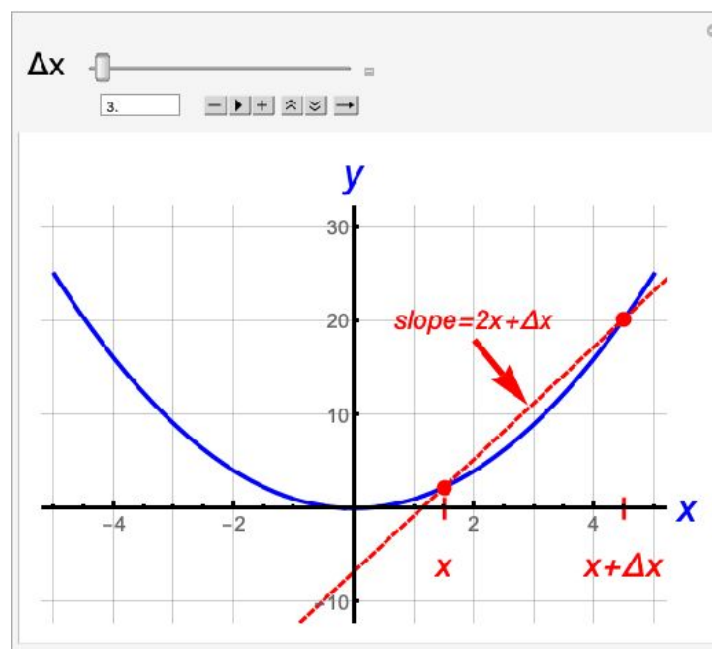


Image source: <https://infinityisreallybig.com/2019/09/20/the-derivative-of-y-x2/>